

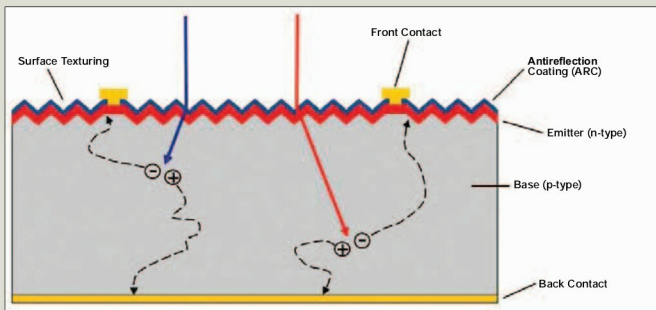


Fig. 1: A solar cell design

Need for simulating solar/PV cells

Since the structural simplicity of most first-generation solar cells made them conducive to one-dimensional simulation, some simplified programs targeting solar applications also emerged and were instrumental in establishing a solid foundation for simulation within the photovoltaic community. Solar cell is designed to convert as many incident photons into electrical current as possible. In order to improve this performance, gradual refinements of the first generation of silicon solar cells eventually led to designs with surfaces employing texturing and anti-reflective coatings to minimise light reflection across the solar spectrum, as shown in Fig. 1.

The base layer of this design is around 250 micrometres thick. Structural and process variables such as the pitch of the front contact, minority carrier lifetime, doping of the base layer, and surface recombination velocities of front and back contacts have significant impact on cell performance. These variables have been subjected to extensive simulation studies, with optimised designs achieving 15 to 16 per cent conversion efficiency.

One of the major critical factors limiting the performance of these cells is the very thick base layer, which provides the opportunity for photon-generated carriers to re-combine before reaching the contacts.

One of the key design aspects is the optimal spacing and size of the rear point contact—a problem that has been well addressed with 3D simulation. Compared to traditional stripe contacts, the current that flows into point contacts has a pronounced 3D character. 3D simulation shows a steep efficiency versus pitch relation for point contacts compared to stripe contacts, which are less sensitive to pitch. This means that simulation is important in order to find the optimal efficiency as a function of contact pitch in point-contacted cells.

As the complexity of solar cell designs has evolved, so has the need for more and more sophisticated 2D and 3D simulation tools to help engineers optimise the cell design in order to achieve the required performance and cost target. Purely experimental approaches are no longer sufficient for solar cell development because of the large number of structural and process variables at play, and through the detailed and often highly non-

linear ways, these variables affect cell performance.

Once a cell design is optimised, the next step would be to combine these cells into reliable, efficient and manufacturable modules. At the module level, the behaviour of the cell can be simplified to a one- or a two-diode model, including the resistive shunt and series connections between the individual cells. Lastly, it is impractical to design new cells without simulation as too many experiments are needed to investigate the requisite design space.

Major design challenges

The major challenges while designing a solar/PV system are:

Minimising interconnect losses. Interconnect losses are very common in solar panels apart from being the most common form of power leakage. Minimising these losses becomes a priority, especially when the losses start mounting and affecting performance.

Evaluating and minimising the effects of environmental variation. The success/failure of solar cells depends on a host of fickle parameters like light intensity, incidence angle of the sun rays, temperature variation and the ionic state of the envi-